

THE MATHEMATICS OF SPNO

Spectral Identifiability and Structure-Preserving Neural Operators for the Parametric Nonlinear Schrödinger Equation

A derivation-level reference for the ten-phase study
organised by subject, with a phase map in §13

Every claim in this document is labelled by *kind*, following the house rule of the project: a **Theorem** or **Proposition** is a mathematical statement with a proof; an **Architectural guarantee** holds for every parameter value θ , untrained weights included, and is asserted in the test suite at random initialisation; a **Numerical observation** is something that was measured and could have come out otherwise. The distinction is load-bearing. “Conserves energy” is never written where the truth is “phase-only substeps preserve the discrete L^2 norm by construction, and Hamiltonian agreement is empirical.”

Numerical constants quoted in the text are the measured values from configuration hash `bd4e108527` ($N = 64$, $\Delta t = 0.01$, $M = 32$ substeps) unless stated otherwise, and are collected in §14.

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How to read this

The study asks one question: *does encoding physical structure directly in a neural operator’s architecture beat an unconstrained baseline, or softer alternatives?* Four ways of using physics are kept rigorously distinct, because they have different mathematical status:

nothing	Model A, a plain FNO. No invariant is preserved; §3.
constraint projection	Model B, FNO composed with a metric projection onto the mass sphere. One scalar invariant, exactly; §4.
soft physics loss	Model P (PINO), a Crank–Nicolson residual penalty. Nothing is guaranteed — and the residual itself smuggles in mass conservation; §8.
architecture	Models C1/C2/C3, learned split-step operators. Mass, reversibility, symplecticity and $U(1)$ equivariance hold by construction; §5.

Part I builds the continuum and discrete theory the whole project rests on. Part II derives what each model class guarantees and, equally important, what it does not. Part III is the identifiability theory — the thesis spine — which is where the only genuinely *negative* theorem lives. Part IV covers the baselines and the two misspecification dials that turn “structure beats FNO” from a tautology into a question. Part V collects measurement machinery and the phase map.

Sections are cross-referenced to the source file that implements them, in a rule like this one:

IN CODE: `src/spno/equations/nls.py`

Part I Foundations

1 The equation, its Hamiltonian structure, and its invariants

1.1 Statement and sign conventions

The object of study is the *parametric* nonlinear Schrödinger equation on the periodic box $\mathbb{T}^d = [0, 2\pi)^d$ (in practice $d = 1$):

$$\boxed{i\psi_t + \alpha\Delta\psi + \beta|\psi|^2\psi - V(x)\psi = 0} \quad \psi(\cdot, t) : \mathbb{T}^d \rightarrow \mathbb{C}, \quad V : \mathbb{T}^d \rightarrow \mathbb{R}. \quad (1)$$

Equivalently, solved for the time derivative,

$$\psi_t = i(\alpha\Delta\psi + \beta|\psi|^2\psi - V\psi). \quad (2)$$

The parameters (α, β) are drawn per sample; $\alpha > 0$ throughout, and β takes either sign (focusing $\beta > 0$, defocusing $\beta < 0$). The learning target is the one-step solution operator

$$\Phi_{\Delta t} : (\psi_n, V, \alpha, \beta) \mapsto \psi_{n+1} = \psi(\cdot, t_n + \Delta t).$$

Three sign conventions follow from (2) and are used without further comment. They are stated here because a single flipped sign invalidates every downstream number, and each has been pinned by a test rather than by a comment.

1. **Kinetic Fourier multiplier** $e^{-i\alpha|k|^2\tau}$.
2. **Local physical-space phase** $e^{+i(\beta|\psi|^2 - V)\tau}$.
3. **Plane-wave dispersion** $\omega = \alpha|k|^2 - \beta|A|^2 + V_0$.

1.2 Hamiltonian structure

Definition 1.1 (Energy).

$$\mathcal{H}[\psi] = \int_{\mathbb{T}^d} \left(\alpha |\nabla\psi|^2 + V|\psi|^2 - \frac{\beta}{2} |\psi|^4 \right) dx.$$

Treating ψ and $\bar{\psi}$ as independent, (1) is the Hamiltonian system

$$i\psi_t = \frac{\delta\mathcal{H}}{\delta\bar{\psi}}. \quad (3)$$

Proposition 1.2. *Definition 1.1 generates (1) through (3).*

Proof. Write $\mathcal{H} = \int \alpha \nabla\psi \cdot \nabla\bar{\psi} + V\psi\bar{\psi} - \frac{\beta}{2}(\psi\bar{\psi})^2$. Varying in $\bar{\psi}$ and integrating the gradient term by parts (no boundary term, the domain is periodic),

$$\frac{\delta\mathcal{H}}{\delta\bar{\psi}} = -\alpha\Delta\psi + V\psi - \beta|\psi|^2\psi.$$

Substituting into (3) gives $i\psi_t + \alpha\Delta\psi + \beta|\psi|^2\psi - V\psi = 0$, which is (1). ■

Remark 1.3. The sign of the quartic term in $\mathcal{H}$ is fixed by this computation and is easy to get wrong: $-\frac{\beta}{2}|\psi|^4$, not $+$. The implementation writes the energy density as $\alpha|\nabla\psi|^2 + V\rho - \frac{\beta}{2}\rho^2$ with $\rho = |\psi|^2$.

1.3 The two conserved quantities

Definition 1.4 (Mass and momentum).

$$\mathcal{M}[\psi] = \int_{\mathbb{T}^d} |\psi|^2 dx, \quad \mathcal{P}[\psi] = \text{Im} \int_{\mathbb{T}^d} \bar{\psi} \nabla \psi dx.$$

Theorem 1.5 (Mass conservation, via Noether). *$\mathcal{H}$ is invariant under the global gauge rotation $\psi \mapsto e^{ic} \psi$, $c \in \mathbb{R}$. Consequently $\mathcal{M}$ is conserved along solutions of (1), for every V , α and β .*

Proof. Invariance is immediate: every term of $\mathcal{H}$ involves ψ paired with $\bar{\psi}$, so the phases cancel. For conservation, compute directly:

$$\frac{d}{dt} \mathcal{M} = \int 2 \text{Re}(\bar{\psi} \psi_t) = \int 2 \text{Re}(\bar{\psi} (\alpha \Delta \psi + \beta |\psi|^2 \psi - V \psi)).$$

The terms $\beta |\psi|^4$ and $-V |\psi|^2$ are real, so i times them is purely imaginary and contributes nothing. For the remaining term, integrate by parts: $\int \bar{\psi} \Delta \psi = - \int |\nabla \psi|^2$, which is real, so $\text{Re}(i \alpha \cdot (\text{real})) = 0$. Hence $d\mathcal{M}/dt = 0$. ■

Theorem 1.6 (Momentum conservation, only when $V \equiv 0$). *If $V \equiv 0$ then (1) is invariant under translations and $\mathcal{P}$ is conserved. For non-constant V it is not.*

Proof. With $V \equiv 0$ the equation has no explicit x -dependence, so $\psi(\cdot, t) \mapsto \psi(\cdot - a, t)$ maps solutions to solutions; Noether’s theorem for the translation group gives conservation of $\mathcal{P}$. Concretely, in $d = 1$,

$$\frac{d\mathcal{P}}{dt} = \text{Im} \int (\bar{\psi}_t \psi_x + \bar{\psi} \psi_{xt}) = - \int V_x |\psi|^2,$$

after substituting (2) and integrating by parts; the kinetic and nonlinear terms produce exact derivatives, which integrate to zero on $\mathbb{T}$. The right-hand side vanishes for every ψ precisely when $V_x \equiv 0$. ■

Proposition 1.7 (The Strang step conserves momentum too, when $V \equiv 0$). *On the periodic grid with spectral differentiation, $\Phi_{\Delta t}$ preserves the discrete momentum $\mathcal{P}_h[\psi] = \mu \text{Im} \sum_j \bar{\psi}_j (\partial_x^{\text{spec}} \psi)_j$ exactly when $V \equiv 0$.*

Proof. In Fourier, $\mathcal{P}_h = \frac{\mu}{N} \sum_k k |\hat{\psi}_k|^2$, which the kinetic substep leaves invariant because it changes only the *phases* of $\hat{\psi}_k$. For the local substep, write $\psi \mapsto e^{i\vartheta} \psi$ with $\vartheta = \beta \rho \Delta t$ real. Then

$$\overline{e^{i\vartheta} \psi} \partial_x (e^{i\vartheta} \psi) = \bar{\psi} \partial_x \psi + i \vartheta_x |\psi|^2,$$

so the momentum picks up $\int \vartheta_x \rho = \beta \Delta t \int \rho_x \rho = \frac{\beta \Delta t}{2} \int (\rho^2)_x = 0$, the integral of a derivative over a periodic domain — which vanishes discretely as well, since the $k = 0$ coefficient of a spectral derivative is exactly zero. ■

Numerical observation 1.8 (Verified). Over 200 steps at $\Delta t = 0.01$ in `float64` on a random band-limited field: relative momentum drift 6.7×10^{-10} with $V \equiv 0$ (roundoff, accumulating linearly from 2.6×10^{-13} after one step) versus 6.4×10^{-2} with $V = 0.4 \cos 2x$. Eight orders of magnitude of separation.

Remark 1.9 (Why this matters for the experiments). The shift arm `G3-potential-zero` sets $V \equiv 0$, which restores translation invariance and therefore adds a *third* exactly conserved quantity to the problem that is absent in the training distribution. That arm is consequently not just “an easier potential”: it is a structurally different problem, and it is the one arm where a model could be right for a reason none of the training data exercised.

1.4 Plane waves and the dispersion relation

Proposition 1.10 (Plane wave). *For constant potential $V \equiv V_0$, the field*

$$\psi(x, t) = A e^{i(k \cdot x - \omega t)}, \quad A > 0, k \in \mathbb{Z}^d,$$

solves (1) exactly if and only if

$$\omega = \alpha |k|^2 - \beta A^2 + V_0. \quad (4)$$

Proof. Substituting: $\psi_t = -i\omega\psi$, $\Delta\psi = -|k|^2\psi$, $|\psi|^2 = A^2$ (constant in x and t). Then (1) reads $\omega\psi - \alpha|k|^2\psi + \beta A^2\psi - V_0\psi = 0$, and dividing by $\psi \neq 0$ gives (4). The “only if” direction is the same computation read backwards. ■

Equation (4) is the ground truth every trained model — black-box models included — is probed against in Part III. Note the three separate dependencies it encodes: quadratic in k , linear in α , and linear in the *amplitude* through βA^2 . The last one is why the probe amplitude cannot be chosen freely (§6.2).

Proposition 1.11 (Plane-wave mass). $\mathcal{M}[Ae^{ik \cdot x}] = A^2 \prod_j L_j$, which for $\mathbb{T}^d = [0, 2\pi)^d$ is $A^2(2\pi)^d$.

1.5 The discrete setting

All computation happens on a uniform, endpoint-free grid of N points per dimension, with spacing $h_j = L_j/N_j$ and cell volume $\mu = \prod_j h_j$. The discrete Fourier transform uses the `norm="backward"` convention: the forward transform carries no factor and the inverse carries $1/N$.

Definition 1.12 (Discrete objects). For a grid field $\psi \in \mathbb{C}^N$:

$$\mathcal{M}_h[\psi] = \mu \sum_j |\psi_j|^2, \quad \hat{\psi}_k = \sum_j \psi_j e^{-ikx_j}, \quad \Delta_h \psi = \mathcal{F}^{-1}(-|k|^2 \hat{\psi}),$$

with wave vectors $k = 2\pi \text{fftfreq}(N, h)$, so that on $[0, 2\pi)$ they are exactly the integers $\{0, 1, \dots, N/2 - 1, -N/2, \dots, -1\}$.

Lemma 1.13 (Discrete Parseval). $\sum_j |\psi_j|^2 = \frac{1}{N} \sum_k |\hat{\psi}_k|^2$. Consequently, multiplying $\hat{\psi}$ pointwise by any function of modulus one preserves $\mathcal{M}_h$.

Proof. Standard for the unnormalised DFT: $\sum_k |\hat{\psi}_k|^2 = \sum_k \sum_{j,l} \psi_j \bar{\psi}_l e^{-ik(x_j - x_l)} = N \sum_j |\psi_j|^2$, using $\sum_k e^{-ik(x_j - x_l)} = N \delta_{jl}$. The second claim follows because $|m_k \hat{\psi}_k| = |\hat{\psi}_k|$ when $|m_k| = 1$. ■

Lemma 1.13 is the entire reason the kinetic substep of every operator in this project conserves mass exactly, and it is why the mass guarantee is *architectural* rather than merely accurate.

Trap 1.14 (Wave vectors must be built in `float64` and narrowed, not built natively). At $N = 64$, evaluating $2\pi \text{fftfreq}$ in `float64` lands 2.3×10^{-13} from the exact integers and narrows to them exactly; evaluating the same expression in `float32` lands 1.2×10^{-4} away, and that error *survives* a later widening to `float64`. Nine orders of magnitude above the 10^{-13} bounds the structural tests assert. Every $|k|^2$ buffer in the codebase is therefore constructed in double and then narrowed — including on rebinding to a new grid.

Trap 1.15 (A float32 parameter poisons a float64 field). `torch.tensor([0.9])` is float32. Feeding it as α to a complex128 field costs $\approx 2.6 \times 10^{-8}$ in α , which moves the one-step phase at $k = 30$ by $\approx 5 \times 10^{-7}$ — invalidating every 10^{-9} assertion in the project. `batch_parameter` rejects the downcast rather than promoting silently.

IN CODE: `src/spno/equations/nls.py`, `src/spno/domain.py`

2 Operator splitting

2.1 The two sub-flows

Write (2) as $\psi_t = (\mathcal{A} + \mathcal{B})\psi$ with

$$\mathcal{A}\psi = i\alpha\Delta\psi \quad (\text{kinetic, linear, diagonal in Fourier}),$$

$$\mathcal{B}\psi = i(\beta|\psi|^2 - V)\psi \quad (\text{local, nonlinear, diagonal in physical space}).$$

Each generator has a *closed-form exact flow*. That is the whole point of the splitting: neither piece is approximated.

Lemma 2.1 (Exact kinetic flow). *The solution of $\psi_t = i\alpha\Delta\psi$ is $\mathcal{K}_\tau\psi = \mathcal{F}^{-1}\left(e^{-i\alpha|k|^2\tau}\hat{\psi}\right)$.*

Proof. In Fourier, Δ acts as multiplication by $-|k|^2$, so $\partial_t\hat{\psi}_k = -i\alpha|k|^2\hat{\psi}_k$, a scalar linear ODE per mode with solution $\hat{\psi}_k(\tau) = e^{-i\alpha|k|^2\tau}\hat{\psi}_k(0)$. On the grid this is exact for every mode the grid represents; there is no spatial discretisation error. ■

Lemma 2.2 (Exact local flow; the density is frozen). *Along $\psi_t = i(\beta|\psi|^2 - V)\psi$, the density $\rho = |\psi|^2$ is constant in time pointwise. Consequently the flow is $\mathcal{N}_\tau\psi = e^{i(\beta\rho - V)\tau}\psi$, with ρ evaluated at the initial state.*

Proof. $\partial_t\rho = 2\operatorname{Re}(\bar{\psi}\psi_t) = 2\operatorname{Re}(i(\beta\rho - V)|\psi|^2) = 0$, because $\beta\rho - V$ is real and $i \times (\text{real})$ is purely imaginary. With ρ frozen, the equation becomes $\psi_t = ic(x)\psi$ with c independent of t , whose solution is the stated multiplier. ■

Lemma 2.2 is the single most important structural fact in the project. It is what makes the local substep exactly invertible, exactly mass-preserving, and exactly Hamiltonian — and it is what breaks in Model C3, where the phase is allowed to read $\operatorname{Re}\psi$ and $\operatorname{Im}\psi$ separately (§5.5).

2.2 The Strang step

Definition 2.3 (Symmetric / Strang splitting).

$$\Phi_{\Delta t} = \mathcal{K}_{\Delta t/2} \circ \mathcal{N}_{\Delta t} \circ \mathcal{K}_{\Delta t/2}$$

The palindromic ordering is not cosmetic. It buys three separate properties — second order, time-symmetry, and (with symplectic substeps) a symmetric symplectic integrator — none of which the asymmetric Lie–Trotter step $\mathcal{K}_{\Delta t}\mathcal{N}_{\Delta t}$ has.

Theorem 2.4 (Exact mass conservation). $\mathcal{M}_h[\Phi_{\Delta t}\psi] = \mathcal{M}_h[\psi]$ exactly, in exact arithmetic, for every $\Delta t, \alpha, \beta, V$.

Proof. $\mathcal{N}$ multiplies each grid value by $e^{i(\beta\rho - V)\Delta t}$, a complex number of modulus one (the exponent is real by Lemma 2.2), so $|\psi_j|$ is unchanged at every j and hence $\sum_j |\psi_j|^2$ is unchanged. $\mathcal{K}$ multiplies each Fourier coefficient by $e^{-i\alpha|k|^2\tau}$, again of modulus one, so by Lemma 1.13 $\sum_j |\psi_j|^2$ is again unchanged. A composition of norm-preserving maps preserves the norm. ■

Theorem 2.5 (Exact time-reversibility). $\Phi_{-\Delta t} \circ \Phi_{\Delta t} = \operatorname{Id}$ exactly.

Proof. Expand, writing $h = \Delta t/2$:

$$\Phi_{-\Delta t}\Phi_{\Delta t} = \mathcal{K}_{-h}\mathcal{N}_{-\Delta t}\mathcal{K}_{-h}\mathcal{K}_h\mathcal{N}_{\Delta t}\mathcal{K}_h.$$

The inner pair cancels: $\mathcal{K}_{-h}\mathcal{K}_h = \text{Id}$, since the Fourier multipliers $e^{+i\alpha|k|^2h}$ and $e^{-i\alpha|k|^2h}$ multiply to 1 mode by mode. It remains to show $\mathcal{N}_{-\Delta t}\mathcal{N}_{\Delta t} = \text{Id}$. Let w be the state entering $\mathcal{N}_{\Delta t}$ and put $\rho_w = |w|^2$. Then $v := \mathcal{N}_{\Delta t}w = e^{i(\beta\rho_w - V)\Delta t}w$, and by Lemma 2.2 $|v|^2 = \rho_w$ — the backward step reads the *same* density field. Hence $\mathcal{N}_{-\Delta t}v = e^{-i(\beta\rho_w - V)\Delta t}v = w$. The outer pair $\mathcal{K}_{-h} \cdots \mathcal{K}_h$ then cancels likewise. ■

Theorem 2.6 (Symplecticity). $\Phi_{\Delta t}$ is a symplectic map on the phase space $(\psi, \bar{\psi})$ with the standard structure induced by (3).

Proof. By Lemma 2.1, $\mathcal{K}_\tau$ is the exact time- τ flow of the Hamiltonian $\mathcal{H}_K = \alpha \int |\nabla\psi|^2$; by Lemma 2.2, $\mathcal{N}_\tau$ is the exact time- τ flow of $\mathcal{H}_N = \int (V\rho - \frac{\beta}{2}\rho^2)$. (Both identifications are verified in the general learned setting in Theorem 5.2(3), of which this is the special case $\kappa = -\alpha|k|^2$, $\nu = \beta\rho - V$.) The exact flow of a Hamiltonian is symplectic, and a composition of symplectic maps is symplectic. ■

Theorem 2.7 (Second order). $\Phi_{\Delta t}$ is consistent of order 2: its local error is $O(\Delta t^3)$.

Proof. Two routes; both are worth having.

(i) *Symmetry.* By Theorem 2.5 the method satisfies $\Phi_{-\Delta t} \circ \Phi_{\Delta t} = \text{Id}$. Write the local error as $\Phi_{\Delta t} = \varphi_{\Delta t} + \sum_{p \geq p_0} \Delta t^{p+1} E_p$ where φ is the exact flow. Time-symmetry forces the expansion of Φ in Δt to contain only odd powers beyond the exact flow, so the leading error exponent $p_0 + 1$ is odd, i.e. the order p_0 is even. Since the method is consistent ($p_0 \geq 1$), $p_0 \geq 2$.

(ii) *Baker–Campbell–Hausdorff.* For generators $\mathcal{A}, \mathcal{B}$,

$$e^{\frac{h}{2}\mathcal{A}}e^{h\mathcal{B}}e^{\frac{h}{2}\mathcal{A}} = \exp\left(h(\mathcal{A} + \mathcal{B}) + h^3\mathcal{C}_2 + O(h^5)\right),$$

$$\mathcal{C}_2 = -\frac{1}{24}[\mathcal{A}, [\mathcal{A}, \mathcal{B}]] + \frac{1}{12}[\mathcal{B}, [\mathcal{B}, \mathcal{A}]]. \quad (5)$$

The h^2 term vanishes identically by the palindromic structure — this is exactly why the asymmetric Lie–Trotter step is only first order — leaving a local error $O(h^3)$. ■

Corollary 2.8 (Modified Hamiltonian and bounded energy error). $\Phi_{\Delta t}$ is the exact flow of a modified Hamiltonian $\tilde{\mathcal{H}} = \mathcal{H} + \Delta t^2\mathcal{H}_2 + O(\Delta t^4)$, where $\Delta t^2\mathcal{H}_2$ is built from the double commutators in (5). Consequently the error in $\mathcal{H}$ along a numerical trajectory is $O(\Delta t^2)$ and **bounded**, oscillating about a level rather than growing with the horizon.

This is the falsifiable prediction that the whole rollout-diagnostics apparatus (§12.2) exists to test. It is what separates a structure-preserving integrator from a merely accurate one: an accurate non-symplectic method has nothing preventing *secular* energy drift.

Numerical observation 2.9 (The reference solver behaves as predicted). Over 1000 steps at $\Delta t = 0.01$ in float64: maximum relative mass drift 3.82×10^{-13} ; maximum relative energy drift 7.92×10^{-5} , with the late-window maximum $0.83 \times$ the early-window maximum — classified *bounded*. Measured Strang convergence order, finest ratio: 2.0465.

Trap 2.10 (Pre-asymptotic convergence ratios are meaningless). The measured order ratios across substeps (4, 8, 16, 32, 64) are (1.69, 4.24, 2.22, 2.05). The 4.24 is a cancellation artefact in the pre-asymptotic regime, not a fourth-order method. Only the finest ratio is quoted, and convergence claims are always made by *measured scaling* ($\log_2(e_{i+1}/e_i)$ compared against the expected order) rather than by an absolute threshold.

2.3 The substepped reference, and why targets cannot be single Strang steps

Definition 2.11 (Substepped reference).

$$\Phi_{\Delta t}^{\text{ref}} = (\Phi_{\Delta t/M})^{\circ M}, \quad M = 32.$$

Remark 2.12 (The central methodological confound). This is not a numerical nicety; it is what stops the study from being circular. Model C is, by construction, *a single Strang step with learned rates*. If the training targets were produced by a single Strang step at the same Δt , the true one-step map would lie *inside* Model C’s hypothesis class, and “structure beats FNO” would be a tautology — C would win because it was handed the answer, not because structure helps. Targets must therefore come from a converged, substepped reference, and every error is quoted relative to it.

Definition 2.13 (The splitting floor).

$$\varepsilon_{\text{split}} = \|\Phi_{\Delta t} - \Phi_{\Delta t}^{\text{ref}}\|_{\text{rel } L^2},$$

the relative L^2 error of one Strang step at Δt against the substepped reference, measured on the production distribution.

$\varepsilon_{\text{split}}$ is an *analytic lower bound on the one-step error of any model whose hypothesis class is a single split step at Δt with the exact rates*. Its role in the argument is precise:

- A structured model that trains *down to* $\varepsilon_{\text{split}}$ has recovered the physics — there is nothing left in its class to find.
- A black-box model that goes *below* $\varepsilon_{\text{split}}$ is exploiting non-split structure that the splitting cannot represent. That is a finding about expressivity, not a failure of the bound.
- A black-box model that sits well *above* $\varepsilon_{\text{split}}$ has left headroom for the structured models. This is what was measured: the FNO is $11\times$ above the floor.

2.4 The fitted floor: the bound a *learned* split step actually faces

Definition 2.13 measures the error of the Strang map built from the *true* rates $\kappa = -\alpha|k|^2$, $\nu = \beta\rho - V$. A learned split step is under no obligation to use those. It may pick *effective* rates that absorb part of the $O(\Delta t^2)$ commutator, so the quantity that genuinely lower-bounds Model C is

$$\varepsilon_{\text{split}}^{\text{fit}} = \inf_{\theta} \|\Phi_{\Delta t}^{\theta} - \Phi_{\Delta t}^{\text{ref}}\|. \quad (6)$$

Two nested families are fitted, both shared across the whole batch (a single model must serve every (α, β)):

$$(\text{scalar}) \quad \kappa = -c_0 \alpha |k|^2, \quad \nu = c_1(\beta\rho - V); \quad (\text{per-mode}) \quad \kappa = -\alpha |k|^2 g(|k|^2),$$

with g a free vector initialised at 1. The per-mode family is exactly the κ_2 parameterisation Phase 4 gives Model C1, so its achieved error is the target C1-K2 must be measured against. Note that g *multiplies* $\alpha|k|^2$ rather than replacing it: a free function of $|k|^2$ alone could not represent the required α -scaling.

Numerical observation 2.14 ($\varepsilon_{\text{split}}$ is a genuine bound). Fitting the effective generators recovers the exact rates to within 10^{-4} ($c_0 = 1.0000008$, $c_1 = 0.9999979$) and improves the error by at most $\sim 5\%$: $\varepsilon_{\text{split}}^{\text{fit}}/\varepsilon_{\text{split}} = 0.9999978$ for the scalar family and 0.99999 for the free 32-parameter per-mode family. *The Strang commutator is very nearly orthogonal to the span of the generators.* There is no “cheap” effective rate that absorbs it.

Numerical observation 2.15 (Reference self-convergence). $M=32$ versus $M=64$ differ by 1.50×10^{-7} , i.e. 0.24% of $\varepsilon_{\text{split}}$. The reference is converged well below the quantity it is used to measure.

IN CODE: `src/spno/solvers/split_step.py`

Part II What each model class guarantees

3 Model A — the unrestricted Fourier neural operator

3.1 The spectral convolution

Definition 3.1 (Mode-truncated global convolution). For a hidden state $u \in \mathbb{R}^{C \times N}$ and learned complex weights $R \in \mathbb{C}^{C \times C' \times m}$,

$$(Su) = \mathcal{F}^{-1} \left(\mathbf{1}_{|k| < m} \cdot R_k \hat{u}_k \right),$$

where $\mathcal{F}$ is the real FFT and m is the retained mode budget (`modes=16`). Modes above m are set to zero, not merely damped.

An FNO layer is $u \mapsto \sigma(Su + Wu)$ with W a pointwise (1×1) convolution. The full model lifts 5 real input channels $(\text{Re } \psi, \text{Im } \psi, V, \alpha, \beta)$ to width 64, applies four such layers, and projects to 2 output channels read as $(\text{Re } \psi', \text{Im } \psi')$.

Proposition 3.2 (Grid independence, and its exact scope). *The spectral weights R are indexed by frequency, and the inverse transform receives the target length n at call time. Hence the same weights define the same operator on any grid — **for $|k| \leq m$ only**. They say nothing whatever about modes above the budget.*

This is the *only* source of the FNO’s much-advertised resolution transfer, and §11 is largely about not overclaiming it.

Architectural guarantee 3.3 (Model A guarantees nothing). Φ^A is an arbitrary trained map $\mathbb{C}^N \rightarrow \mathbb{C}^N$. It does not conserve $\mathcal{M}_h$, does not conserve $\mathcal{H}$, is not time-reversible, is not $U(1)$ -equivariant, and does not respect (4). Any agreement with these is learned and approximate.

Remark 3.4 (Δt does not enter the map). The FNO ignores its Δt argument entirely. Consequently: (i) evaluating it at $-\Delta t$ re-applies the *forward* map, so any “reversibility” number measured that way is an artefact — `evaluate_reversibility` refuses rather than returning it; and (ii) it has no basis for Δt -transfer. The base class encodes this as `supports_time_reversal` and `supports_dt_transfer` flags that raise rather than silently returning a wrong answer.

3.2 Three design choices, each tied to a confound

1. **Parameter conditioning is rescaled to $O(1)$.** α and β are broadcast over the grid and mapped affinely from their sampling ranges to $[-1, 1]$:

$$\tilde{\alpha} = \frac{2(\alpha - \alpha_{\min})}{\alpha_{\max} - \alpha_{\min}} - 1.$$

A consequence worth recording: with $\alpha \in [0.7, 1.1]$, the value $\alpha = 0$ enters the lift channel at $\tilde{\alpha} = -4.5$. Any α -continuation experiment (§6.6) that anchors near $\alpha = 0$ is therefore probing α -extrapolation, and a failure there says nothing about whether the α -derivative information was extracted.

2. **No absolute-coordinate channel.** The joint operator $(\psi, V) \mapsto \psi'$ is translation-equivariant — the potential, not the coordinate, is what breaks homogeneity. Feeding x into the lift lets the model memorise position instead of learning the operator. Kept behind a flag as an ablation.

3. **A single global amplitude scale, not per-sample normalisation.** The nonlinearity $\beta|\psi|^2\psi$ is *not* scale-invariant, and mass varies across samples by design (§10). Normalising each sample by its own norm would destroy exactly the information the nonlinear term depends on.

3.3 The three separable causes of high- k failure

When an FNO does badly at large $|k|$, three mechanisms are in play, and only one of them is interesting. Keeping them apart is the entire design of experiment G7.

- (a) **Hard mode truncation beyond m .** The spectral branch contributes nothing above $|k| = m = 16$. High- k content can only reach the output through the pointwise branches and the nonlinearity. *Trivial*, and removable by matching m to Nyquist.
- (b) **The oscillatory dependence of $e^{-i\alpha|k|^2\Delta t}$ on α .** The FNO must route this through a pointwise lift of a broadcast scalar. *This is the interesting one* — it is a statement about inductive bias.
- (c) **No training energy above k_{train} .** Initial conditions are band-limited to $|k| \leq 8$; nothing above that is supervised. *No architecture fixes this.*

IN CODE: `src/spno/models/fno.py`

4 Model B — hard constraint projection

Definition 4.1 (Mass projection).

$$\Pi_M(\psi) = \sqrt{\frac{M}{\mathcal{M}_h[\psi]}} \psi, \quad \Phi_{\Delta t}^B(\psi_n) = \Pi_{\mathcal{M}_h[\psi_n]}(\Phi_{\Delta t}^A(\psi_n)).$$

Proposition 4.2 (Π_M is the metric projection onto the mass sphere). *Let $S_R = \{u : \|u\|_{L_h^2} = R\}$ with $R = \sqrt{M}$. For $\psi \neq 0$,*

$$\Pi_M(\psi) = \arg \min_{u \in S_R} \|u - \psi\|_{L_h^2}, \quad \text{and} \quad \min_{u \in S_R} \|u - \psi\| = |R - \|\psi\||.$$

Proof. For $u \in S_R$, $\|u - \psi\|^2 = R^2 + \|\psi\|^2 - 2\operatorname{Re}\langle u, \psi \rangle$, so minimising the distance maximises $\operatorname{Re}\langle u, \psi \rangle$. By Cauchy–Schwarz, $\operatorname{Re}\langle u, \psi \rangle \leq \|u\|\|\psi\| = R\|\psi\|$ with equality iff u is a positive multiple of ψ ; the unique such u in S_R is $R\psi/\|\psi\|$, which is $\Pi_M(\psi)$. Substituting gives $\|u - \psi\|^2 = (R - \|\psi\|)^2$. ■

So the name is earned: this is genuinely the closest mass-correct field, not merely a convenient rescaling.

Architectural guarantee 4.3 (Exact mass, and nothing else). $\mathcal{M}_h[\Phi_{\Delta t}^B\psi_n] = \mathcal{M}_h[\psi_n]$ exactly in exact arithmetic, for every parameter value. Model B guarantees **no other** property: not phase, not energy, not reversibility, not the dispersion relation.

Remark 4.4 (Two training modes, and they are not equivalent). *Projection in the loop* makes Π part of the forward pass, so gradients flow through it and the core learns to produce fields whose *shape* is right with amplitude handled downstream. *Post hoc* trains the core unconstrained and applies Π only at evaluation, so the core still spends capacity on the amplitude. They must be reported separately.

Numerical observation 4.5 (What one hard invariant actually buys). Across 3 seeds at matched budget: the projection removes mass drift across **13 orders of magnitude** ($1.61 \times 10^{-2} \rightarrow 1.3 \times 10^{-15}$ at step 100) and buys **~12% rollout accuracy** ($2.77 \times 10^{-2} \rightarrow 2.44 \times 10^{-2}$), of which ~ 7 points come from the projection alone and ~ 5 from training with it. Energy remains **secular** for A, B-post and B-loop alike, in 9/9 seed-model runs. *Projecting mass does not fix energy*. If that reads as a disappointing result, it is the intended lesson: one scalar constraint is one scalar constraint.

IN CODE: `src/spno/models/projected.py`, `src/spno/domain.py`

5 Model C — structure encoded in the architecture

5.1 The parameterisation

Definition 5.1 (Learned split step).

$$\Phi_{\Delta t}^\theta = \mathcal{K}_{\Delta t/2}^\theta \circ \mathcal{N}_{\Delta t}^\theta \circ \mathcal{K}_{\Delta t/2}^\theta$$

with

$$\widehat{\mathcal{K}_\tau^\theta \psi}(k) = e^{i\tau \kappa_\theta(|k|^2, \alpha, \beta)} \widehat{\psi}(k), \quad \mathcal{N}_\tau^\theta \psi = e^{i\tau \nu_\theta(\rho, V, \alpha, \beta)} \psi, \quad \rho = |\psi|^2,$$

and **both κ_θ and ν_θ real-valued**. The true values are $\kappa = -\alpha|k|^2$ and $\nu = \beta\rho - V$.

Everything that follows depends on exactly two facts: the rates are *real*, and ν_θ sees ψ *only through ρ* .

5.2 The main theorem

Theorem 5.2 (Structure of the learned split step). *For every θ — untrained weights included — the map $\Phi_{\Delta t}^\theta$ of Definition 5.1 satisfies:*

- (1) **Exact mass.** $\mathcal{M}_h[\Phi^\theta \psi] = \mathcal{M}_h[\psi]$.
- (2) **Exact reversibility.** $\Phi_{-\Delta t}^\theta \circ \Phi_{\Delta t}^\theta = \text{Id}$.
- (3) **Symplecticity, as the Strang splitting of a learned Hamiltonian.** There exists $\mathcal{H}_\theta = \mathcal{H}_{K,\theta} + \mathcal{H}_{N,\theta}$ with

$$\mathcal{H}_{K,\theta} = - \int \kappa_\theta(|k|^2) |\widehat{\psi}(k)|^2, \quad \mathcal{H}_{N,\theta} = - \int F_\theta(\rho, V), \quad \frac{\partial F_\theta}{\partial \rho} = \nu_\theta,$$

such that each substep is the exact flow of its sub-Hamiltonian. Hence $\Phi_{\Delta t}^\theta$ is symplectic and time-symmetric.

- (4) **Global $U(1)$ equivariance.** $\Phi^\theta(e^{ic} \psi) = e^{ic} \Phi^\theta(\psi)$ for every $c \in \mathbb{R}$.

Proof.

(1). ν_θ is real-valued by construction, so $e^{i\Delta t \nu_\theta}$ has modulus one pointwise and $|\psi_j|$ is unchanged at every grid point. κ_θ is real-valued, so $e^{i\tau \kappa_\theta}$ has modulus one mode by mode and Lemma 1.13 applies. Composition of norm-preserving maps.

(2). The kinetic multipliers cancel in pairs exactly as in Theorem 2.5, because κ_θ does not depend on ψ at all — it is a function of $(|k|^2, \alpha, \beta)$ only, so the forward and backward half-steps use *identical* rates. For the local step, let w be its input, $\rho_w = |w|^2$, and $v = e^{i\Delta t \nu_\theta(\rho_w, \cdot)} w$. By (1) applied pointwise, $|v|^2 = \rho_w$, so ν_θ evaluated at v equals ν_θ evaluated at w — the backward step reads the same phase field — and $e^{-i\Delta t \nu_\theta(\rho_w, \cdot)} v = w$. This is the step where ρ -only dependence is used, and it is the step that fails for C3.

(3). Consider first the kinetic generator. The flow $\partial_t \hat{\psi}_k = i\kappa_k \hat{\psi}_k$ is Hamiltonian with $\mathcal{H}_{K,\theta} = -\sum_k \kappa_k |\hat{\psi}_k|^2$ (up to the DFT normalisation), since

$$\frac{\partial \mathcal{H}_{K,\theta}}{\partial \hat{\psi}_k} = -\kappa_k \hat{\psi}_k \implies i \partial_t \hat{\psi}_k = \frac{\partial \mathcal{H}_{K,\theta}}{\partial \hat{\psi}_k} \iff \partial_t \hat{\psi}_k = i\kappa_k \hat{\psi}_k.$$

$\mathcal{H}_{K,\theta}$ is real because κ_θ is.

Now the local generator $\partial_t \psi = i\nu_\theta(\rho, V)\psi$. Fix (V, α, β) and define

$$F_\theta(\rho) = \int_0^\rho \nu_\theta(s, V, \alpha, \beta) ds,$$

which exists because ν_θ is a continuous (indeed smooth) function of the *scalar* ρ — this is the point at which the pointwise structure is indispensable, and it is why an antiderivative always exists no matter what the network has learned. Set $\mathcal{H}_{N,\theta} = -\int F_\theta(\rho) dx$. Using $\rho = \psi \bar{\psi}$ so that $\partial \rho / \partial \bar{\psi} = \psi$,

$$\frac{\delta \mathcal{H}_{N,\theta}}{\delta \bar{\psi}} = -\frac{\partial F_\theta}{\partial \rho} \cdot \frac{\partial \rho}{\partial \bar{\psi}} = -\nu_\theta \psi \implies i \partial_t \psi = -\nu_\theta \psi \cdot (-1) \iff \partial_t \psi = i\nu_\theta \psi,$$

as required. Finally, by the argument of Lemma 2.2 (which used only that ν_θ is real), ρ is constant along this flow, so the closed-form multiplier $e^{i\tau \nu_\theta}$ is the exact time- τ flow — not an approximation to it. Each substep is therefore an exact Hamiltonian flow, hence symplectic; $\Phi_{\Delta t}^\theta$ is a composition of symplectic maps, and is time-symmetric by the palindromic ordering together with (2).

(4). $\mathcal{K}^\theta$ is linear, so it commutes with multiplication by the constant e^{ic} . And ρ is phase-blind: $|e^{ic}\psi|^2 = |\psi|^2$. Hence ν_θ is unchanged and $\mathcal{N}^\theta$, being multiplication by a ψ -independent (given ρ) pointwise factor, also commutes with the global phase. Composing, $\Phi^\theta(e^{ic}\psi) = e^{ic}\Phi^\theta(\psi)$. ■

Remark 5.3 (Two routes to (1)). Part (1) also follows from (3) and (4) by Noether's theorem: $\mathcal{H}_\theta$ depends on ψ only through $|\hat{\psi}|^2$ and ρ , so it is $U(1)$ -invariant, and the conserved charge of that symmetry is $\mathcal{M}_h$. The direct modulus-one argument is the one the tests assert, because it holds mode by mode and grid point by grid point and is therefore checkable to machine precision.

5.3 The sharp, falsifiable prediction

Part (3) is not decoration. It is the claim the study can be wrong about.

Corollary 5.4 (Bounded, not secular, energy error). Φ^θ does not conserve the true Hamiltonian $\mathcal{H}$. What it does is integrate a learned Hamiltonian $\mathcal{H}_\theta$ exactly-symplectically. By backward error analysis (Corollary 2.8 applied to $\mathcal{H}_\theta$), the drift in $\mathcal{H}_\theta$ is bounded; the drift in $\mathcal{H}$ is therefore governed by the fixed model-error term $\|\mathcal{H}_\theta - \mathcal{H}\|$ evaluated along the trajectory. That is a constant of the model, not a function of the horizon. **Prediction: energy drift is bounded and oscillatory, not secular.**

Phases 2–3 measured *secular* energy drift for the FNO and for the mass-projected FNO in 9/9 seed–model runs (Observation 4.5), so this prediction is falsifiable against a real baseline rather than against a straw man.

Remark 5.5 (The reversible-KAM caveat — state it before the result comes in). Symmetric and reversible methods already enjoy bounded-energy results for reversible systems, independently of symplecticity. Model C2 is reversible but not symplectic (§5.4). It is therefore **unlikely that C1 and C2 separate on energy drift**, and a finding that they do not is not evidence against Corollary 5.4. Frame the C1-vs-C2 comparison as *sample efficiency versus expressivity*, not as a test of symplecticity.

5.4 C2 — what survives a nonlocal phase

Model C2 replaces the pointwise v_θ with an FNO reading the four real channels (ρ, V, α, β) . The phase is now a *nonlocal* functional of ρ .

Architectural guarantee 5.6 (C2). Parts (1), (2) and (4) of Theorem 5.2 hold verbatim. Part (3) does not.

Proof. (1) and (4) used only that v_θ is real and phase-blind, both of which survive. (2) used that ρ is unchanged by the local step — and it is unchanged *as an entire field*, not merely at the point being updated. A nonlocal v reads the whole ρ field, which is frozen, so the backward step still sees the identical phase field and still cancels exactly.

For (3), the local generator $\partial_t \psi = i\nu[\rho]\psi$ is Hamiltonian iff ν is the functional derivative of some $G[\rho]$, i.e. iff its linearisation is self-adjoint:

$$\frac{\delta \nu(x)}{\delta \rho(y)} = \frac{\delta \nu(y)}{\delta \rho(x)} \quad \text{for all } x, y.$$

This is a codimension-large symmetry condition on the operator, and an unconstrained FNO does not satisfy it. Hence C2’s local substep is generically not the flow of a sub-Hamiltonian and the composition is not symplectic. ■

Remark 5.7 (C2 is the fair fight). Model A and Model C2 share the *same backbone*, the same hyperparameters and essentially the same parameter count (287,746 vs 288,770, a ratio of $1.004\times$). They differ in exactly one thing: whether the network’s output is the field, or is a *phase applied* to the field. That is the cleanest available isolation of “structure” from “capacity”. C1, at 2,466 parameters, is $117\times$ smaller than A — which is why parameter counts must appear in every table.

5.5 C3 — the control, and why “irreversible” would be the wrong word

Model C3 feeds $\text{Re } \psi$ and $\text{Im } \psi$ to the phase network instead of ρ .

Architectural guarantee 5.8 (C3). Part (1) of Theorem 5.2 still holds — the multiplier is still modulus one. Part (4) is destroyed outright. Part (2) degrades from *exact* to *second order*.

Proof of the reversibility statement. Write the local step as $u \mapsto v = e^{i\Delta t \nu(u)} u$, where now ν depends on u and not merely on $|u|$. The backward step gives

$$\mathcal{N}_{-\Delta t}(v) = e^{-i\Delta t \nu(v)} v = e^{i\Delta t [\nu(u) - \nu(v)]} u.$$

Now $v = u + O(\Delta t)$, indeed $v = u + i\Delta t \nu(u) u + O(\Delta t^2)$, so by smoothness of ν ,

$$\nu(v) - \nu(u) = D\nu[u](i\Delta t \nu(u) u) + O(\Delta t^2) = O(\Delta t).$$

Hence the residual phase is $\Delta t \cdot O(\Delta t) = O(\Delta t^2)$, and it is generically nonzero, because $D\nu[u]$ applied to $i\nu(u)u$ — a purely “rotational” perturbation — vanishes only if ν happens to be phase-blind, which is exactly what C3 gives up. Since the kinetic halves are unitary and cancel exactly, the round-trip error of the full step is $O(\Delta t^2)$. ■

Numerical observation 5.9 (Measured order 2.00). At untrained weights, C3’s forward-backward violation is $\approx 1 \times 10^{-9}$ at $\Delta t = 0.01$, against $\approx 7 \times 10^{-16}$ and Δt -independent for C1/C2. Its measured Δt -scaling is order 2.00.

Trap 5.10 (Absolute thresholds hide power laws). 10^{-9} is small enough that any fixed tolerance would call C3 “reversible”. It is not; it is second-order. Regimes are therefore classified by *measured Δt -scaling* — $\log_2(e_{i+1}/e_i)$ compared against an expected order — never by an absolute threshold. The defensible claim is:

ρ -only parameter sharing makes reversibility exact rather than second-order

and *not* “C3 is irreversible”.

C3 has two jobs. It is the direct control for Model B — both preserve mass and nothing else, at very different capacities — and it is the **paired negative** proving the C1/C2 reversibility and $U(1)$ tests are not vacuous. A conservation test that would pass on a model with no such structure tests nothing.

5.6 The rate ladders: approximation versus optimisation

Both learned rates are, in truth, *products*: $\kappa = -\alpha \cdot |k|^2$ and $\nu = \beta \cdot \rho - V$. A tanh MLP handed $(|k|^2, \alpha, \beta)$ or (ρ, V) as features must *discover* the bilinear form. Whether it can is a question about learning, not about expressivity, and the two must not be conflated. Hence two ladders:

rung	kinetic κ	what is handed to the network
K0	$\text{MLP}(\tilde{k}^2, \alpha, \beta)$	nothing — must find the product
K1	$\text{MLP}(\tilde{k}^2, \alpha k^2, \alpha, \beta)$	the product as an extra feature
K2	$-\alpha k ^2 (1 + \text{MLP}(\cdot))$	the functional form; MLP only corrects
rung	local ν	
L0	$\text{MLP}(\rho, V; \alpha, \beta)$	nothing
L1	$\text{MLP}(\rho, V, \beta\rho; \alpha, \beta)$	the product $\beta\rho$
L2	$\beta\rho - V + \text{MLP}(\cdot)$	the exact rate; MLP only corrects

K2 and L2 are zero-initialised at the last layer, so they *start* at the exact rate.

Remark 5.11 (Why k_0/L_0 are the headline). A win above k_{wrap} at k_2 is uninterpretable: the model was told the dispersion relation. A win at k_0 means it *learned* it. The ladder is reported in full, because if L_0 cannot reach the floor that is itself a finding — about how hard the learning problem is, distinct from how expressive the class is.

Numerical observation 5.12 (The approximation floor of a free local rate). Measured on the production distribution, a free MLP plateaus at 2.1% relative RMSE on v (a wider, deeper one at 2.6% — an *optimisation* limit, not a capacity one). End-to-end, with the kinetic rate held exact:

$$L_0 : 1.18 \varepsilon_{\text{split}}, \quad L_1 : 1.03 \varepsilon_{\text{split}}, \quad L_2 : 0.93 \varepsilon_{\text{split}}.$$

So a free local rate inflates $C1$'s one-step error by $\sim 18\%$, *not* by the factor of ~ 3 that a naive $\Delta t \cdot |v| \cdot e$ bound predicts — the phase error is partly incoherent across the grid and does not accumulate as that bound assumes. L_2 landing slightly *below* the floor is expected, and matches Observation 2.14: a free correction to the generators absorbs a few percent of the Strang commutator.

Trap 5.13 (Normalise $|k|^2$ before it reaches a tanh). At $N = 64$, $|k|^2$ reaches 1024 and saturates every tanh in the network. The kinetic MLP therefore consumes $\tilde{k}^2 = |k|^2 / \max |k|^2$. The normaliser is **frozen at the training grid** when the model is rebound to a finer grid — see Proposition 11.3.

5.7 Summary of guarantees

	mass	reversibility	symplectic	$U(1)$
A (FNO)	—	—	—	—
B (FNO + projection)	exact	—	—	—
C1 (pointwise ρ phase)	exact	exact	exact	exact
C2 (FNO-on- ρ phase)	exact	exact	—	exact
C3 (Re/Im ψ phase)	exact	order 2	—	—
reference (Strang)	exact	exact	exact	exact

Every “exact” in this table is asserted in the test suite **at random untrained weights**, each with a paired negative demonstrating the test can fail. Positives bound the worst case ($\max < 10^{-13}$); negatives bound the best case ($\min > 10^{-6}$).

IN CODE: `src/spno/models/split_learned.py`

Part III Spectral identifiability — the thesis spine

6 The one-step multiplier and what it can tell you

The headline question of the study is *architecture-relative identifiability*:

Single- α supervision makes $\omega(k)$ unrecoverable above $k_{\text{wrap}} = \sqrt{\pi/(\alpha\Delta t)}$ for **every** model — a clean theorem. α -varying supervision makes it recoverable **in principle**, because $\partial_\alpha \arg m(k, \alpha) = -k^2\Delta t$ is wrap-free. The question is which hypothesis classes actually extract it. *The data contains it; only the right inductive bias gets it out.*

This section proves both halves.

6.1 The probe

Definition 6.1 (One-step multiplier). Fix a constant potential $V \equiv V_0$, an amplitude A , and parameters (α, β) . For any one-step map Φ (a trained model or the reference solver), let

$$m(k) = \frac{\widehat{\Phi(\psi)}(k)}{\widehat{\psi}(k)}, \quad \psi = A e^{ik \cdot x}.$$

Proposition 6.2 (The splitting is *exact* on plane waves in a constant potential). For $V \equiv V_0$ and $\psi = A e^{ik \cdot x}$, the single Strang step, the substepped reference, and the exact flow all coincide:

$$\Phi_{\Delta t} \psi = \Phi_{\Delta t}^{\text{ref}} \psi = e^{-i\omega\Delta t} \psi, \quad \omega = \alpha|k|^2 - \beta A^2 + V_0.$$

Proof. $\mathcal{K}_{\Delta t/2} \psi = e^{-i\alpha|k|^2\Delta t/2} \psi$, a constant multiple of ψ , so the midpoint state has $\rho \equiv A^2$, constant in x . The local phase $e^{i(\beta A^2 - V_0)\Delta t}$ is therefore also a constant multiple. Composing the three constants gives $e^{-i(\alpha|k|^2 - \beta A^2 + V_0)\Delta t}$. The same argument applies to each substep of Φ^{ref} , and the constants compose to the same total. By Proposition 1.10 this is the exact flow. ■

Remark 6.3 (The floor for probe residuals is roundoff, not $\varepsilon_{\text{split}}$). Proposition 6.2 has a sharp consequence that is easy to get wrong: $\varepsilon_{\text{split}} = 6.239 \times 10^{-5}$ was measured on *random band-limited fields*, where ρ varies in x and the commutator is nonzero. It is **not** the floor for dispersion-probe residuals. On plane waves in a constant potential the floor is `float64` roundoff, $\sim 10^{-13}$ in ω at $\Delta t = 0.01$, and the probe self-validation gate is set accordingly at 10^{-10} .

Remark 6.4 (What the probe is *not*). Three limits, each load-bearing and each stated in the thesis rather than buried:

- (i) Plane waves are **out of distribution** for every model here — all are trained on random band-limited fields. This probes the *learned operator*, not generalisation.
- (ii) The probe is valid for **constant** V_0 **only**, because that is the only case in which a plane wave is an exact solution.
- (iii) The branch lift (§6.3) selects the branch *using the truth*. Below k_{wrap} the branch index is zero and the lift is a consistency check; above it the lift is an assist and the “recovered” ω is not an independent measurement. That asymmetry is the G5a result, not a defect in the instrument.

6.2 The probe amplitude must not smuggle in a mass shift

By (4), ω depends on the amplitude through βA^2 ; by Proposition 1.11, the plane wave's mass is $A^2 \prod_j L_j$. A probe amplitude outside the training mass distribution therefore makes every high- k reading a measurement of *two* shifts at once — spectral and mass — with no way to attribute the failure.

Definition 6.5 (Probe amplitude). With the training mass range $[M_-, M_+]$ and $\bar{M} = \frac{1}{2}(M_- + M_+)$,

$$A = \sqrt{\bar{M} / \prod_j L_j}.$$

For the production configuration, $M \in [1, 3]$ and $\prod_j L_j = 2\pi$, so $A = \sqrt{2/(2\pi)} \approx 0.5642$.

6.3 The negative theorem

Theorem 6.6 (Single- α ambiguity). *Let the supervision consist of one-step data at a **fixed** α and fixed Δt . Then for each mode k , the data determine $\omega(k)$ only modulo $2\pi/\Delta t$. Writing the principal branch*

$$\omega_{\text{princ}}(k) = -\frac{\arg m(k)}{\Delta t} \in \left[-\frac{\pi}{\Delta t}, \frac{\pi}{\Delta t}\right),$$

one has $\omega_{\text{princ}}(k) = \omega(k)$ if and only if $|\omega(k)|\Delta t \leq \pi$. In particular, with the kinetic term dominant, the horizon is

$$k_{\text{wrap}} = \sqrt{\frac{\pi}{\alpha \Delta t}} \quad (7)$$

*and above it the branch is **genuinely unrecoverable** from the data — for every model, of every architecture.*

Proof. The only thing a one-step map on a plane wave reveals about mode k is the complex number $m(k)$, and by Proposition 6.2 the ground truth is $m(k) = e^{-i\omega(k)\Delta t}$. The map $\omega \mapsto e^{-i\omega\Delta t}$ is $2\pi/\Delta t$ -periodic, so $m(k)$ determines $\omega(k)$ exactly up to that period. The principal-branch statement is the definition of $\arg$ taking values in $(-\pi, \pi]$. Setting $|\omega|\Delta t = \pi$ with $\omega \approx \alpha k^2$ and solving for k gives (7). ■

Corollary 6.7 (Multi-step data at the same Δt adds nothing). *Let $m_n(k)$ be the n -step multiplier. Then $m_n = m^n$, so m_n is a deterministic function of m : the n -step observation is measurable with respect to the σ -algebra generated by the one-step observation and carries no additional information about ω . (Conversely m_n determines m only up to an n -th root, so it is strictly weaker.)*

Proof. A plane wave in a constant potential remains a plane wave of the same amplitude under the flow (Proposition 6.2), so iterating multiplies the coefficient by m each time. ■

6.4 The alias family, constructed

Theorem 6.6 has a constructive shadow that makes it testable.

Proposition 6.8 (Indistinguishable α values at a given mode). *Fix k and Δt . Define the alias family*

$$\alpha_n = \alpha_0 + \frac{2\pi n}{k^2 \Delta t}, \quad n \in \mathbb{Z}. \quad (8)$$

*Then $m(k; \alpha_n) = m(k; \alpha_0)$ for every n : the one-step map restricted to mode k is **identical** across the family.*

Proof. $\arg m(k; \alpha) = -(\alpha k^2 - \beta A^2 + V_0)\Delta t$. Replacing α_0 by $\alpha_0 + 2\pi n / (k^2 \Delta t)$ changes the argument by $-2\pi n$, i.e. not at all as a complex number. ■

Numerical observation 6.9 (The computed family at $k = 30$). With $\Delta t = 0.01$, $k = 30$: $2\pi / (k^2 \Delta t) = 2\pi / 9 = 0.69813$, so from $\alpha_0 = 0.9$ the family runs $\{0.9, 0.20187, -0.49627, \dots\}$. The test asserts a bit-identical map across these values, with the offsets *computed*, never hardcoded — the spec’s rounded literals (0.2019, -0.4963) carry $\sim 7 \times 10^{-5}$ of residual α , which moves the phase at $k = 30$ far above the assertion tolerance.

Remark 6.10 (The ambiguity is per-mode, and that is the right statement). No single $\alpha' \neq \alpha$ aliases at *every* k simultaneously, since (8) demands $\alpha' - \alpha = 2\pi n_k / (k^2 \Delta t)$ with n_k integer for all k at once, which fails already for two coprime k . So Theorem 6.6 is not a claim that the scalar α is unidentifiable — a model that has the form $\omega = \alpha k^2 + \dots$ baked in can pin α down from low k and then predict high k correctly. It is a claim about **nonparametric recovery of the curve** $\omega(k)$: an unconstrained model has no reason to prefer one branch at large k , because every branch fits the data exactly.

This is precisely why the theorem is interesting for a study about inductive bias, and precisely why k_0 rather than k_2 is the headline rung. k_2 is handed the functional form and therefore escapes the ambiguity by assumption; k_0 must discover it.

6.5 The positive theorem: the α -derivative is wrap-free

Theorem 6.11 (α -derivative identifiability). *For every k , including $k > k_{\text{wrap}}$,*

$$\frac{\partial}{\partial \alpha} \arg m(k, \alpha) = -k^2 \Delta t, \quad (9)$$

with β and V_0 dropping out entirely. Moreover the derivative can be estimated from wrapped data without any unwrapping: for a grid $\alpha_0 < \alpha_1 < \dots$ with gaps $\Delta \alpha_j$,

$$\arg(m_{j+1} \overline{m_j}) = -\Delta \alpha_j k^2 \Delta t \quad \text{exactly, provided} \quad \boxed{|\Delta \alpha_j| k^2 \Delta t < \pi}$$

Proof. From $\arg m(k, \alpha) = -(\alpha k^2 - \beta A^2 + V_0)\Delta t$ (as a real-valued lift), differentiating in α gives (9); the β and V_0 terms are α -independent and vanish.

For the estimator: $m_{j+1} \overline{m_j}$ is a complex number whose true argument is $-\Delta \alpha_j k^2 \Delta t$. The function $\arg$ returns the representative in $(-\pi, \pi]$, so it returns *exactly* that value if and only if $|\Delta \alpha_j k^2 \Delta t| \leq \pi$. Thus consecutive wrapped increments equal the true increments under the stated condition, and no phase unwrapping — and in particular no reference to the truth — is required. ■

Theorem 6.11 is why identifiability survives above k_{wrap} under α -varying supervision even though the *absolute* phase does not. The information is present in the data. Whether a given hypothesis class extracts it is an empirical question — experiment **G5b**, the central result of the study.

Remark 6.12 (The precondition is checked, not assumed). The estimator’s precondition is verified on the caller’s own α grid before any measurement is taken, and the function raises otherwise. The dataset’s realised gap is $\Delta \alpha_{\text{max}} = 3.467 \times 10^{-3}$; the requirement at $k = 32$ is $\Delta \alpha < \pi / (k^2 \Delta t) = \pi / 10.24 = 0.3068$. The margin is a factor of ~ 88 . A *probe* that chooses its own coarser spacing, however, can silently violate it and return garbage — hence the check.

6.6 α -continuation is a different claim

Accumulating wrapped increments from an anchor gives the *absolute* frequency:

$$\arg m(k, \alpha_J) \stackrel{\text{lift}}{=} \arg m(k, \alpha_0) + \sum_{j=0}^{J-1} \arg(m_{j+1} \overline{m_j}), \quad \omega = -\frac{(\cdot)}{\Delta t}.$$

Proposition 6.13 (Anchor condition). *This recovers the true $\omega(k, \alpha_J)$ provided (i) the gap condition of Theorem 6.11 holds on every step, and (ii) the anchor is itself unwrapped: $|\omega(k, \alpha_0)| \Delta t \leq \pi$.*

Condition (ii) forces the anchor toward $\alpha_0 \approx 0$ for large k — far outside the training range $[0.7, 1.1]$. As noted in §3, $\alpha = 0$ enters an FNO’s lift channel at $\tilde{\alpha} = -4.5$. **A failure of α -continuation is therefore evidence about α -extrapolation, not about whether the α -derivative information was extracted**, and the two must be reported as separate claims. Theorem 6.11 deliberately requires *in-range* α only.

6.7 Multi- Δt as a second route past the horizon

Proposition 6.14 (Two step sizes). *Observing the one-step multiplier at step sizes $\Delta t_1, \Delta t_2$ determines ω modulo the generator of the lattice $\frac{2\pi}{\Delta t_1} \mathbb{Z} \cap \frac{2\pi}{\Delta t_2} \mathbb{Z}$, i.e. modulo $\text{lcm}(\frac{2\pi}{\Delta t_1}, \frac{2\pi}{\Delta t_2})$.*

Corollary 6.15 (A dyadic ladder buys only its smallest step). *For $\Delta t \in \{0.005, 0.01, 0.02\}$ the periods are $\{400\pi, 200\pi, 100\pi\}$, each an integer multiple of the next, so their lcm is 400π — the period of the smallest Δt alone. **The two coarser step sizes add no identifiability whatsoever.** The horizon extends only as*

$$k_{\text{wrap}}(\Delta t) = \sqrt{\pi/(\alpha \Delta t)} \propto \Delta t^{-1/2},$$

i.e. by $\sqrt{2}$ per halving: from 16.9 at $\Delta t = 0.01$ to 23.9 at $\Delta t = 0.005$ (for $\alpha = 1.1$).

Remark 6.16. This is worth stating explicitly because it is counterintuitive and cheap to get wrong. If the goal of the multi- Δt arm (G6) were *identifiability*, the step sizes should be chosen *incommensurate* — then the lcm is enormous and the combined horizon far exceeds either alone. As specified, G6’s dyadic ladder is best understood as a test of Δt -**transfer** (does a model trained at one Δt work at another?), which is a different and equally legitimate question; and Δt -transfer is meaningful only for the split-step family, where Δt genuinely multiplies a learned rate. An FNO ignores its Δt argument, so a transfer number measured on one is an artefact (Remark 3.4).

6.8 The horizon is arm-dependent

k_{wrap} is a function of α , so widening the α range *moves* the horizon. On the production configuration $\alpha \in [0.7, 1.1]$, $\Delta t = 0.01$:

$$k_{\text{wrap}} \in [\sqrt{\pi/0.011}, \sqrt{\pi/0.007}] = [16.90, 21.19].$$

The G2 extrapolation arm ($\alpha \in [0.5, 1.5]$) spans $[14.47, 25.07]$. Every plot must therefore draw *its own arm’s* horizon rather than a single global line, and the banded-error edges must be recomputed per arm. The relevant scales for the production configuration are

$$k_{\text{train}} = 8 < k_{\text{wrap}} \in [16.9, 21.2] < k_{\text{Nyq}} = 32.$$

IN CODE: `src/spno/evaluation/dispersion.py`, `src/spno/equations/nls.py`, `src/spno/data/shift.py`

7 Spectral diagnostics

Field-space L^2 hides both failures this study is about. A model can have small relative L^2 while getting the *dispersion relation* wrong — error concentrated in a few high- k modes barely moves the norm — and phase error is invisible to any metric that looks only at $|\psi|$.

7.1 Per-mode and banded error

$$E(k) = \left\langle \frac{|\hat{\psi}_k^{\text{pred}} - \hat{\psi}_k^{\text{true}}|}{|\hat{\psi}_k^{\text{true}}|} \right\rangle_{\text{batch}}, \quad E_{[a,b]} = \left\langle \frac{\|\mathbf{1}_{[a,b]}(\hat{\psi}^{\text{pred}} - \hat{\psi}^{\text{true}})\|}{\|\mathbf{1}_{[a,b]}\hat{\psi}^{\text{true}}\|} \right\rangle.$$

Normalising per mode is what makes low-energy high- k modes visible at all; the absolute version is dominated by whichever modes carry the energy. Default band edges are $k_{\text{train}} = 8$ and $k_{\text{wrap}} = 16.9$.

7.2 Amplitude and phase separate exactly

Proposition 7.1 (Exact amplitude–phase decomposition). *With $p = \hat{\psi}^{\text{pred}}$, $t = \hat{\psi}^{\text{true}}$ and $\Delta\varphi = \arg p - \arg t$,*

$$\underbrace{\sum_k |p_k - t_k|^2}_{\text{total}^2} = \underbrace{\sum_k (|p_k| - |t_k|)^2}_{\text{amplitude}^2} + \underbrace{2 \sum_k |p_k| |t_k| (1 - \cos \Delta\varphi_k)}_{\text{phase}^2},$$

and the phase term is nonnegative, vanishing exactly when the phases agree.

Proof. $|p - t|^2 = |p|^2 + |t|^2 - 2 \operatorname{Re}(p\bar{t}) = (|p| - |t|)^2 + 2|p||t| - 2|p||t| \cos \Delta\varphi$. Sum over k . ■

So taking the phase part as $\sqrt{\text{total}^2 - \text{amplitude}^2}$, as `amplitude_and_phase_split` does, is not a heuristic residual. It is exactly

$$\left(2 \sum_k |p_k| |t_k| (1 - \cos \Delta\varphi_k)\right)^{1/2} \approx \left(\sum_k |p_k| |t_k| \Delta\varphi_k^2\right)^{1/2}$$

for small phase errors.

Trap 7.2 (Never reduce a phase-error curve with `max`). $\Delta\varphi_k$ is meaningful only where the target carries energy. In an *empty* mode the phase is roundoff noise, distributed uniformly on $(-\pi, \pi]$, so a bare $\max_k$ reports $O(1)$ regardless of model quality. The scalar summary must be the amplitude-weighted mean *across modes*,

$$\overline{\Delta\varphi} = \frac{\sum_k |t_k| |\Delta\varphi_k|}{\sum_k |t_k|}.$$

7.3 The G7 normalisation

Fixing α makes the one-step task easier at *every* k , so absolute $E(k)$ falls across the board and an absolute comparison cannot separate the three causes of high- k failure listed in §3.3. Only the ratio isolates cause (b):

$$\text{band ratio} = \frac{E_{[\text{high}, \infty)}}{E_{[0, \text{low}]}} ,$$

and even then only once cause (a) is removed by matching m to Nyquist. This is a *numerical observation* on a trained model; it says nothing at untrained weights.

7.4 The nonlinear cascade

Initial conditions are band-limited to $|k| \leq 8$ *exactly* (a sharp cutoff, not a Gaussian roll-off — see §10), so every mode above the cutoff starts empty and any energy found there is genuine nonlinear transfer through $\beta|\psi|^2\psi$. A model that gets one-step L^2 right while failing to cascade is a failure that field-space L^2 cannot see.

Numerical observation 7.3 (The cascade is real but stays below the horizon). On the training split: energy fraction above $k_{\text{train}} = 8$ grows from 8.3×10^{-6} at step 1 to 2.7×10^{-4} at step 200. Above k_{wrap} it stays at $\sim 5 \times 10^{-7}$ to 2.4×10^{-6} throughout. So in-distribution rollouts do *not* silently become spectral-extrapolation runs — which is what makes G1 a clean control. Measured, not assumed.

IN CODE: `src/spno/evaluation/spectral.py`, `src/spno/data/generate.py`

Part IV The soft baseline and the misspecification dials

8 The PINO residual — and the invariant it smuggles in

8.1 The discrete midpoint residual

Definition 8.1 (Crank–Nicolson / midpoint residual). With $\bar{\psi} = \frac{1}{2}(\psi_{n+1} + \psi_n)$ and $\bar{\rho} = \frac{1}{2}(|\psi_{n+1}|^2 + |\psi_n|^2)$,

$$R = i \frac{\psi_{n+1} - \psi_n}{\Delta t} + \alpha \Delta_h \bar{\psi} + \beta \bar{\rho} \bar{\psi} - V \bar{\psi}.$$

The Laplacian is spectral, so it is exact for every mode the grid represents.

The nonlinear term is averaged in the *Delfour–Fortin–Payre* form $\bar{\rho} \cdot \bar{\psi}$ rather than, say, $|\bar{\psi}|^2 \bar{\psi}$. That choice is what makes the scheme discretely mass-conserving — and it is exactly the problem.

8.2 The confound, proved

Theorem 8.2 (Cayley transform: the CN update has modulus exactly one). *Let $\psi_n = Ae^{ikx}$ with constant $V \equiv V_0$, and let $\psi_{n+1} = z\psi_n$ with $|z| = 1$ (so that $\bar{\rho} = A^2$). Then $R \equiv 0$ if and only if*

$$z = \frac{1 - i\omega\Delta t/2}{1 + i\omega\Delta t/2}, \quad \omega = \alpha k^2 - \beta A^2 + V_0, \quad (10)$$

and this z satisfies $|z| = 1$ **exactly**, for every Δt and every ω .

Proof. With $\psi_{n+1} = z\psi_n$, $\bar{\psi} = \frac{1+z}{2}\psi_n$, $\Delta_h \psi_n = -k^2 \psi_n$ and $\bar{\rho} = A^2$:

$$\frac{R}{\psi_n} = i \frac{z-1}{\Delta t} + (-\alpha k^2 + \beta A^2 - V_0) \frac{1+z}{2} = i \frac{z-1}{\Delta t} - \omega \frac{1+z}{2}.$$

Setting this to zero and writing $a = \omega\Delta t/2$ gives $i(z-1) = a(1+z)$, hence $z(i-a) = a+i$ and

$$z = \frac{a+i}{i-a} = \frac{1-ia}{1+ia},$$

which is (10). Its modulus is $|1-ia|/|1+ia| = \sqrt{1+a^2}/\sqrt{1+a^2} = 1$. ■

Corollary 8.3 (The physics loss contains the invariant under study). *On plane waves the CN residual is exactly solved by a unitary update. More generally the DFP-averaged scheme conserves the discrete mass. The 7a physics loss therefore smuggles in the very invariant the study compares methods on.*

This is not a reason to avoid the baseline — it is precisely what the loss/projection/architecture distinction exists to expose — but it must be stated in the thesis *next to every 7a number*.

Corollary 8.4 (The scheme’s own frequency). *Writing $z = e^{-i\omega_{\text{CN}}\Delta t}$ and inverting (10),*

$$\omega_{\text{CN}} = \frac{2}{\Delta t} \arctan\left(\frac{\omega\Delta t}{2}\right)$$

Proof. $\arg \frac{1-ia}{1+ia} = -2 \arctan a$ with $a = \omega \Delta t / 2$, so $\omega_{\text{CN}} \Delta t = 2 \arctan(\omega \Delta t / 2)$. ■

Trap 8.5 (A test that feeds the analytic plane wave and expects zero tests the wrong thing). The scheme's own exact solution advances at ω_{CN} , not at ω . The gap between them is the scheme's $O(\Delta t^2)$ truncation error, and a residual test must be built around Corollary 8.4, not around $e^{-i\omega \Delta t}$.

8.3 The residual is second order — with the coefficient derived

Proposition 8.6 (Residual expansion). *Feeding the residual the analytic pair $\psi_{n+1} = e^{-i\omega \Delta t} \psi_n$ gives, with $\theta = \omega \Delta t$,*

$$\frac{R}{\psi_n} = \frac{1}{\Delta t} \left[\frac{1}{12} \theta^3 - \frac{i}{24} \theta^4 - \frac{1}{80} \theta^5 + O(\theta^6) \right],$$

that is

$$\text{Re}\left(\frac{R}{\psi_n}\right) = \frac{\omega^3 \Delta t^2}{12} + O(\Delta t^4), \quad \text{Im}\left(\frac{R}{\psi_n}\right) = -\frac{\omega^4 \Delta t^3}{24} + O(\Delta t^5).$$

The next real term is $-\omega^5 \Delta t^4 / 80$, so the leading coefficient is recovered only up to a relative correction of order $\frac{12}{80} (\omega \Delta t)^2 = 0.15 (\omega \Delta t)^2$.

Proof. From the proof of Theorem 8.2 with $z = e^{-i\theta}$,

$$\frac{R}{\psi_n} = \frac{1}{\Delta t} \left[i(e^{-i\theta} - 1) - \frac{\theta}{2}(1 + e^{-i\theta}) \right].$$

Expand $e^{-i\theta} = 1 - i\theta - \frac{\theta^2}{2} + \frac{i\theta^3}{6} + \frac{\theta^4}{24} - \frac{i\theta^5}{120} + \dots$. Then

$$i(e^{-i\theta} - 1) = \theta - \frac{i\theta^2}{2} - \frac{\theta^3}{6} + \frac{i\theta^4}{24} + \frac{\theta^5}{120},$$

$$\frac{\theta}{2}(1 + e^{-i\theta}) = \theta - \frac{i\theta^2}{2} - \frac{\theta^3}{4} + \frac{i\theta^4}{12} + \frac{\theta^5}{48}.$$

Subtracting, the θ and θ^2 terms cancel identically and

$$\theta^3 \left(-\frac{1}{6} + \frac{1}{4} \right) + i\theta^4 \left(\frac{1}{24} - \frac{1}{12} \right) + \theta^5 \left(\frac{1}{120} - \frac{1}{48} \right) = \frac{\theta^3}{12} - \frac{i\theta^4}{24} - \frac{\theta^5}{80}.$$

Dividing by Δt and substituting $\theta = \omega \Delta t$ gives the stated expansion. ■

8.4 Normalising the loss

$$\mathcal{L}_{\text{phys}} = \left\langle \frac{\|R\|_{L_h^2}}{\|\psi_n\|_{L_h^2} / \Delta t} \right\rangle_{\text{batch}}.$$

The normalisation makes the weight λ comparable across Δt and across mass — both of which vary by design. An *unnormalised* residual weight becomes uninterpretable the moment the multi- Δt arm exists, and would quietly turn the objective into a mass-weighted one. The λ sweep $\{0.01, 0.1, 1, 10\}$ is reported in full, never as a tuned value.

IN CODE: `src/spno/losses/pde_residual.py`

9 Misspecification — moving the truth outside the constrained class

Phase 9 is the only phase that makes “structure beats FNO” a *question* rather than a tautology, because it moves the data-generating equation outside the hypothesis class of the structured models. Two dials, each breaking a different assumption, each recovering exact NLS at zero.

9.1 Dial 1 — nonlocal nonlinearity (breaks locality)

Definition 9.1 (Nonlocal generator). Replace the local rate by

$$\nu = \beta (W_\sigma * \rho) - V, \quad \widehat{W}_\sigma(k) = e^{-\sigma^2 |k|^2 / 2}.$$

Proposition 9.2 (The kernel is a unit-mass symmetric mollifier for every σ). $\widehat{W}_\sigma(0) = 1$ exactly, so $\int W_\sigma = 1$ and the kernel never rescales the mean density; $\widehat{W}_\sigma$ is real and even in k , so W_σ is real and symmetric; and $\sigma \rightarrow 0$ gives $\widehat{W} \rightarrow 1$, i.e. a delta, by construction rather than in a limit that has to be argued.

Architectural guarantee 9.3 (Mass and $U(1)$ survive). ν is real, so the local substep is still a modulus-one multiplier: discrete mass is preserved exactly and global $U(1)$ equivariance survives, for every σ . Model B’s projection therefore remains *correct* under this dial.

Theorem 9.4 (The dynamics remain Hamiltonian). *For symmetric W , the local substep of Definition 9.1 is still the exact flow of a sub-Hamiltonian, namely*

$$\mathcal{H}_N = -\frac{\beta}{2} \iint \rho(x) W(x-y) \rho(y) dx dy + \int V \rho dx.$$

*Hence the composition is still symplectic, and ρ is still frozen along the flow. **What is broken is locality, and nothing else.***

Proof. Let $G[\rho] = \frac{\beta}{2} \iint \rho(x) W(x-y) \rho(y)$. Then

$$\frac{\delta G}{\delta \rho(x)} = \frac{\beta}{2} \left(\int W(x-y) \rho(y) dy + \int \rho(y) W(y-x) dy \right) = \beta (W * \rho)(x),$$

where the two terms combine into one *only because* $W(x-y) = W(y-x)$. So $\delta(-G + \int V \rho) / \delta \rho = -[\beta(W * \rho) - V] = -\nu$, and by the computation in the proof of Theorem 5.2(3), $\delta \mathcal{H}_N / \delta \psi = (\delta \mathcal{H}_N / \delta \rho) \psi = -\nu \psi$, which generates $\partial_t \psi = i\nu \psi$. Since ν is real, ρ is frozen (Lemma 2.2) and the closed-form multiplier is the exact flow. ■

Remark 9.5 (Why this dial separates C1 from C2). C1’s ν_θ is *pointwise* in ρ and therefore **cannot represent** $\beta(W_\sigma * \rho)$ for $\sigma > 0$: the truth has left C1’s class. C2’s ν_θ is an FNO on ρ and *can*. This is the cleanest available separation of the two, and it is why the dial was chosen.

Remark 9.6 (Deliberately *not* a saturable or quintic nonlinearity). ν_θ is already a free function of the scalar ρ , so the C family learns $\nu = \beta\rho/(1+s\rho)$ or $\nu = \beta\rho + c\rho^2$ easily. Those are not misspecifications at all — they are inside the class. Only breaking *locality* or breaking *conservation* moves the truth out.

9.2 Dial 2 — weak gain/loss (breaks conservation itself)

Definition 9.7 (Gain/loss generator).

$$i\psi_t + \alpha \Delta \psi + \beta |\psi|^2 \psi - V\psi = +i\gamma \psi, \quad \text{i.e.} \quad \psi_t = \gamma \psi + i(\alpha \Delta \psi + \beta |\psi|^2 \psi - V\psi).$$

Theorem 9.8 (Mass grows at a fixed exponential rate). *Along (9.7),*

$$\boxed{\frac{d}{dt} \ln \mathcal{M} = 2\gamma}$$

Proof. $\frac{d}{dt} \int |\psi|^2 = \int 2 \operatorname{Re}(\bar{\psi} \psi_t) = \int 2 \operatorname{Re}(\gamma |\psi|^2) + \int 2 \operatorname{Re}(i\bar{\psi}(\cdots))$. The second integral vanishes by the computation in Theorem 1.5. The first is $2\gamma \mathcal{M}$. Hence $d\mathcal{M}/dt = 2\gamma \mathcal{M}$. ■

Remark 9.9 (The sign convention is pinned by a measured rate, not by prose). Writing $-i\gamma\psi$ on the right-hand side would give decay instead. The tests assert $d \ln \mathcal{M} / dt = 2\gamma$ as a *rate* rather than trusting the derivation — which is the correct level of paranoia for a sign that is this easy to invert.

Remark 9.10 (This dial makes Model B actively wrong). Under $\gamma \neq 0$ the true mass is *not* conserved, so Model B’s hard projection is no longer merely unhelpful — it enforces a false constraint. A located crossover here is direct evidence about *when a hard invariant stops helping*, which is the generalisation the whole study is aiming at. A *bounded* result (“no crossover within the swept range”) is equally publishable.

Proposition 9.11 (What is exactly, and what is only approximately, exact). *The gain factor $e^{\gamma\tau}$ is a real scalar and therefore commutes with the linear Fourier multiplier $\mathcal{K}_\tau$. Consequently splitting the gain away from the kinetic step introduces **no additional splitting error**, and the dial does not contaminate $\varepsilon_{\text{split}}$.*

It does not, however, commute with the nonlinear local step. Along the exact local sub-flow $\psi_t = \gamma\psi + i(\beta\rho - V)\psi$ one has $\rho(\tau) = \rho_0 e^{2\gamma\tau}$, so the exact accumulated phase is

$$\int_0^\tau (\beta\rho_0 e^{2\gamma s} - V) ds = \beta\rho_0 \tau(1 + \gamma\tau) - V\tau + O(\gamma^2\tau^3),$$

whereas freezing ρ at ρ_0 gives $\beta\rho_0\tau - V\tau$. The local substep therefore carries a phase error $\beta\rho_0\gamma\tau^2 + O(\gamma^2\tau^3)$ — second order in the substep and first order in the dial, hence vanishing as $\gamma \rightarrow 0$ and further suppressed by a factor $1/M$ by the substepping.

The precise statement to make in the thesis is therefore: *gain and kinetic commute exactly; the residual $O(\beta\gamma\Delta t^2/M)$ term lives inside the local substep and is suppressed by the substepped reference along with the ordinary Strang commutator.*

9.3 The dial-zero gate

Trap 9.12 (A spectral Gaussian at $\sigma = 0$ is not the identity). $\widehat{W}_0 \equiv 1$ exactly, but $\mathcal{F}^{-1}(\mathcal{F}\rho \cdot 1)$ is **not bitwise** ρ — the FFT round trip is accurate, not exact. Phase 9’s opening gate asserts that the dial-zero generator reproduces the unperturbed solver *bitwise*, so both perturbed operators short-circuit to the plain Strang forward pass at zero, and the exact arm is routed through the *same class* the production shards were generated with. This is load-bearing, not an optimisation: a crossover measured against a subtly different zero point is uninterpretable.

Trap 9.13 (Turn one dial at a time). Two simultaneous perturbations cannot be attributed to either broken assumption. The configuration object refuses the combination outright.

Remark 9.14 (The perturbed generators must be substepped too). Both dials are wrapped in the same $M = 32$ substepping as the unperturbed reference. A single perturbed Strang step at Δt would confound the dial with the $O(\Delta t^2)$ splitting error, and every located crossover would then be partly a measurement of $\varepsilon_{\text{split}}$.

Remark 9.15 (Do not add other equations). NLS was chosen because it bundles nonlinearity, complex fields, spectral structure, Hamiltonian invariants, phase-sensitive error and long-horizon stability into one equation. Adding Burgers, wave or Navier–Stokes before Phase 9 is complete costs the thesis its spine.

IN CODE: `src/spno/solvers/perturbed.py`, `src/spno/misspecification.py`

Part V Data, measurement, and the phase map

10 The training distribution, and why it is shaped this way

Two design choices exist purely to keep later experiments honest.

1. **Sharp spectral cutoff.** Initial conditions are band-limited *exactly* to $|k| \leq k_{\text{train}} = 8$:

$$\widehat{\psi}_0(k) = \mathbf{1}_{|k| \leq k_{\text{train}}} e^{-|k|^2 / (2k_{\text{train}}^2)} \widehat{\xi}(k), \quad \xi \text{ white noise,}$$

with real and imaginary parts drawn independently (the NLS state is a genuine complex field, not a pair of coupled real ones). A *soft* filter that leaked high- k energy into the training set would void the G4 bandwidth-shift design with no visible symptom.

2. **Mass varies across samples** — drawn uniformly in $[1, 3]$, a $3\times$ range. Normalising every initial condition to unit mass would make mass conservation *memorisable*, which flatters the unconstrained baselines and turns Model B's projection into a no-op.

Potentials are smooth, real, and normalised so that $\max |V|$ equals the sampled amplitude, with amplitude, correlation length and functional family controlled *independently* so that the G3 potential arm can vary one at a time.

10.1 The resolution-adequacy screen

Focusing NLS ($\beta > 0$) can steepen a field until the grid stops resolving it, and the resulting error looks exactly like an out-of-distribution failure. The screen measures the spectral tail fraction above $0.75 k_{\text{Nyq}}$ and requires it to stay below 10^{-6} ; trajectories that fail are excluded, or the (α, β) box is narrowed.

Numerical observation 10.1 (The production box is resolved with enormous margin). Maximum tail fraction over 64 screened samples: 2.52×10^{-10} , median 3.55×10^{-13} , against a threshold of 10^{-6} . Zero samples over threshold. Verdict: *resolved*.

11 Resolution transfer, honestly

11.1 Spectral resampling

Proposition 11.1 (Exact resampling of band-limited fields). *Moving a periodic field between grids by truncating or zero-padding its spectrum is **exact** for any field band-limited below both Nyquists. With the `norm="backward"` convention the inverse transform carries the $1/N$, so coefficients must be rescaled by $N_{\text{tgt}}/N_{\text{src}}$ to hold the physical amplitude fixed rather than the discrete one.*

Proof. $\psi(x_j) = \frac{1}{N} \sum_k \widehat{\psi}_k e^{ikx_j}$. For the continuum function to be unchanged on the target grid one needs $\frac{1}{N'} \sum_k \widehat{\psi}'_k e^{ikx} = \frac{1}{N} \sum_k \widehat{\psi}_k e^{ikx}$ for all x , hence $\widehat{\psi}'_k = (N'/N) \widehat{\psi}_k$ on the common modes and zero elsewhere. Exactness follows because a band-limited field has no coefficients outside that common set. ■

Trap 11.2 (The Nyquist mode is self-conjugate, and cannot be upsampled). On an even grid the mode $k = N/2$ is self-conjugate: it represents $\cos(Nx/2)$ with no way to distinguish $+k$ from $-k$. Splitting it between the two target modes on a finer grid is a **choice, not a fact**, and silently making one would inject an arbitrary phase into every downstream number. Upsampling a field with relative energy above 10^{-12} at the source Nyquist is therefore refused rather than resolved by convention.

11.2 Which parameters are grid-independent

parameter group	status
FNO <code>spectral.weight</code>	grid-independent for $ k \leq m$: indexed by frequency, and <code>irfft</code> receives n at call time. This is the only source of the FNO's resolution transfer, and it says nothing about modes above the budget.
FNO lift / local / project	grid-independent: pointwise in the channel axis.
C-family local phase	grid-independent: pointwise in (ρ, V, α, β) , no grid-shaped parameter. (C2's phase FNO likewise indexes by frequency.)
C-family	grid-dependent input range — see Proposition 11.3.
<code>kinetic.k_squared</code>	
Mass projection	grid-independent: rescaling by a mass ratio involves no grid-shaped parameter.

Proposition 11.3 (The frozen normaliser, and its honest cost). *The kinetic MLP consumes $\tilde{k}^2 = |k|^2 / \max_{\text{grid}} |k|^2$. On rebinding to a finer grid the normaliser is **frozen at the training grid's value**. Freezing is necessary: recomputing it would send the same physical k to a different network input (1024 at $N = 64$, 4096 at $N = 128$), silently changing κ everywhere without touching a weight. The cost, which must be reported rather than hidden, is that modes above the training grid's Nyquist then enter at $\tilde{k}^2 > 1$ — outside the input range the MLP ever saw.*

Remark 11.4 (The genuine resolution test is G4 wearing a different hat). Upsampling a *band-limited* field and re-running an FNO barely changes its output, because the FNO only touches $|k| \leq m$ and those modes are unchanged by the resample. **That is not evidence of resolution transfer.** The genuine test is new energy above the training grid's Nyquist — which is spectral extrapolation (G4), not resolution transfer. The two arms are reported separately, with this sentence attached.

IN CODE: `src/spno/evaluation/resolution.py`

12 Measurement: precision, drift classification, and order estimation

12.1 Precision discipline

Training runs in `float32/complex64` for speed. **Anything measuring a conservation law is explicitly widened to `float64/complex128` and evaluated on CPU**, so that arithmetic roundoff never masks — or manufactures — a genuine architectural violation.

Numerical observation 12.1 (The two floors, quoted beside every “exact” claim). Mass drift of the reference over 100 steps:

$$\text{float64, CPU} : 1.87 \times 10^{-14}, \quad \text{float32, CPU} : 1.43 \times 10^{-5}.$$

The `float32` drift accumulates **linearly**, at $\approx 1.2 \times 10^{-7}$ per split step — not as a random walk. That is the signature of a systematic (biased) round-off in the FFT round trip rather than a stochastic one, and it means the drift is predictable from the step count. MPS is $\approx 3\times$ worse than CPU.

Trap 12.2 (Floors must be measured with one split step per model step). The trajectory generator takes $M = 32$ substeps per stored step. Measuring a `float32` mass floor with the generator’s substepping would report $32\times$ the drift a model actually incurs. Use one split step per model step.

Trap 12.3 (Three ways to destroy a complex network while widening it). (i) `.double()` on a complex tensor **silently drops the imaginary part**. (ii) `.double()` on a module **skips** complex parameters entirely. (iii) `.to(torch.float64)` **casts the complex spectral weights to real**, destroying the operator with only a warning. All three are converted into hard exceptions by an explicit dtype guard inside the spectral convolution, and widening goes through a single dedicated helper.

Trap 12.4 (`model.to(device)` mutates in place). It will strand a core that another model shares — and Model B-post deliberately reuses Model A’s weights.

12.2 Bounded versus secular: classification by fitted slope

“Bounded” and “secular” differ by whether drift keeps *growing with the horizon*, which is a statement about a trend, so it is fitted rather than eyeballed:

$$\log d_i \approx c + s \log n_i, \quad \text{classification} = \begin{cases} \text{bounded} & s < 0.25 \\ \text{secular} & s \geq 0.25 \end{cases}$$

with s reported together with its standard error. A conserved-up-to-model-error quantity plateaus ($s \approx 0$); linear accumulation gives $s \approx 1$. The cut at 0.25 is deliberately **generous to “bounded”**, so that a borderline case is reported as secular rather than flattering the structured model. Points with $d_i \leq 0$ are dropped: they are exact conservation, which has no slope.

12.3 Order estimation

Every order claim in the project is a measured $\log_2$ ratio compared against the expected value, never a threshold:

$$p_i = \log_2 \frac{e_{i+1}}{e_i}, \quad \text{regime} = \begin{cases} \text{exact} & \max_i e_i < 10^{-12} \text{ and flat in } \Delta t \\ \text{approximate, order } \bar{p} & \max_i p_i - \min_i p_i < 0.25 \\ \text{inconsistent} & \text{otherwise} \end{cases}$$

Varying Δt away from the trained value is deliberate here: it probes the *architecture*, and is not a claim about prediction accuracy at those step sizes. Because a trained model legitimately refuses off- Δt evaluation (Remark 3.4), the licence to do so is a scoped, temporary opt-in rather than a flag that stays flipped.

12.4 Run-identifier hygiene

Every swept knob enters the run identifier. A four-value λ sweep writing to one directory silently overwrites three results. The pattern is `f"{config_hash(data_config)}-{arm}-{knob}"`,

and the misspecification identifier deliberately *collapses* to the bare data hash at dial zero so the exact arm reuses the existing shards untouched while every perturbed arm gets its own directory.

Gates use `raise RuntimeError`, never `assert`: `python -O` strips asserts, and a gate is the only thing standing between an instrument bug and a full run of plausible-but-fictional numbers.

13 The phase map

Which mathematics each phase of the pipeline rests on.

Ph.	What it does	Mathematics it rests on
0	Reference validation	Lemmas 2.1–2.2 (exact sub-flows); Theorems 2.4–2.7 (mass, reversibility, symplecticity, order 2); Corollary 2.8 (bounded energy); Definition 2.13 and §2.4 (the floor and the fitted floor); §10 (resolution screen).
1	Data generation	§10 (sharp cutoff, varying mass, potential families); Remark 2.12 (targets must be substepped); Observation 7.3 (the cascade stays below k_{wrap}); Theorem 6.11 precondition ($\Delta\alpha k^2\Delta t < \pi$) checked on the realised draw.
2–3	Baselines A and B	§3 (FNO structure, grid independence, the three high- k causes); Proposition 4.2 (projection is the metric projection); Guarantee 4.3; §12.2 (secular vs bounded classification).
4–5	Structured models C1/C2/C3	Theorem 5.2 (the four guarantees, with proofs); Corollary 5.4 (the falsifiable prediction); Guarantees 5.6–5.8 and the C3 order-2 proof (§5.5); Remark 5.5 (reversible-KAM caveat); §5.6 (the K/L ladders and Observation 5.12).
6	Generalisation and identifiability (<i>the spine</i>)	Definition 6.1 and Proposition 6.2 (the probe, and why its floor is roundoff not $\varepsilon_{\text{split}}$); Theorem 6.6 (G5a: the negative theorem) with Corollary 6.7 and Proposition 6.8/Remark 6.10; Theorem 6.11 (G5b: the positive theorem); Proposition 6.13 (why continuation is a <i>different</i> claim); Corollary 6.15 (G6: what a dyadic Δt ladder actually buys); §6.2 (G-arms: probe amplitude and the mass confound); §7 (G7 band ratio; G9 cascade).
7	PINO baseline	Definition 8.1; Theorem 8.2 and Corollary 8.3 (the physics loss contains the invariant under study); Corollary 8.4 (the scheme’s own frequency); Proposition 8.6 (second order, coefficients derived); loss normalisation.
8	Resolution and robustness	Proposition 11.1 and Trap 11.2 (exact resampling; the Nyquist obstruction); Proposition 11.3 (the frozen normaliser and its cost); Proposition 3.2 (what grid independence does and does not mean); §12 (noise, sparsity, and the precision floors).
9	Misspecification sweep	Definition 9.1, Proposition 9.2, Theorem 9.4 (nonlocal but still Hamiltonian — what separates C1 from C2); Definition 9.7 and Theorem 9.8 ($d \ln \mathcal{M} / dt = 2\gamma$); Proposition 9.11 (what is exactly vs approximately exact under the dial); Trap 9.12 (the bitwise gate).

14 Constants and measured values

All at configuration hash `bd4e108527`: $N = 64$, $\Delta t = 0.01$, $M = 32$ substeps, $\alpha \sim U(0.7, 1.1)$, $\beta \sim U(-0.4, 0.6)$, ψ_0 band-limited to $|k| \leq 8$, $\text{mass} \sim U(1, 3)$, 800/100/100 disjoint trajectories $\times 200$ steps = 160,000 one-step training pairs.

quantity	value
$\varepsilon_{\text{split}}$ (splitting floor, mean)	6.239×10^{-5}
median / min / max	$4.484 \times 10^{-5} / 1.155 \times 10^{-6} / 2.801 \times 10^{-4}$
$\varepsilon_{\text{split}}^{\text{fit}} / \varepsilon_{\text{split}}$, scalar family	0.9999978
$\varepsilon_{\text{split}}^{\text{fit}} / \varepsilon_{\text{split}}$, per-mode (k2) family	0.99999
fitted kinetic / local scale	1.0000008 / 0.9999979
Strang convergence order (finest ratio)	2.0465
reference self-convergence, $M=32$ vs $M=64$	1.50×10^{-7} (0.24% of $\varepsilon_{\text{split}}$)
$k_{\text{train}} / k_{\text{wrap}} / k_{\text{Nyq}}$	8 / 16.90–21.19 / 32
α max gap (train)	3.467×10^{-3} (needs < 0.3068)
probe amplitude A	0.5642
mass floor, float64 CPU (100 steps)	1.87×10^{-14}
mass floor, float32 CPU (100 steps)	1.43×10^{-5}
float32 drift per split step	$\approx 1.2 \times 10^{-7}$, linear
long run: max mass drift, float64 (1000 steps)	3.82×10^{-13}
long run: max energy drift; classification	7.92×10^{-5} ; bounded
resolution screen: max tail fraction	2.52×10^{-10} (threshold 10^{-6})

Phases 2–3 (3 seeds, 25 epochs, 287,746 params each)

model	1-step	roll@100	mass@100	energy@100	trend
A (FNO)	6.94×10^{-4}	2.77×10^{-2}	1.61×10^{-2}	1.79×10^{-2}	secular
B-post	6.74×10^{-4}	2.57×10^{-2}	1.3×10^{-15}	9.19×10^{-3}	secular
B-loop	6.62×10^{-4}	2.44×10^{-2}	1.2×10^{-15}	8.14×10^{-3}	secular

The FNO does **not** beat $\varepsilon_{\text{split}}$; it sits $11\times$ above it, so the structured models have headroom.

Phases 4–5 (guarantees verified at *random untrained weights*)

model	params	vs A	mass @init	reversibility
C1 (pointwise ρ phase)	2,466	$0.009\times$	4.4×10^{-16}	exact (Δt -independent)
C2 (FNO-on- ρ phase)	288,770	$1.004\times$	2.2×10^{-16}	exact
C3 (Re/Im ψ phase)	288,834	$1.004\times$	4.4×10^{-16}	order 2.00

A vs C2 is the fair fight: same backbone, matched capacity, differing only in whether the network output is the field or a phase applied to it.

A note on outstanding debt

Phases 2–3 as tabulated were **budget-bound**: `best_epoch == 24` for all six runs (early stopping never fired) with validation loss still falling 26–50% over the last five epochs. The numbers above are therefore a lower bound on all three models’ achievable accuracy and are not yet convergence-comparable. Nothing in this document’s *mathematics* depends on them; the guarantees are architectural and hold at random initialisation. But no accuracy number should enter the thesis until the budget-bound warning is empty for every model in the comparison.